

Department of Mathematics, IIT Madras
MA2031 Linear Algebra July - November 2023
Assignment 1

1. Let $A = [a_{ij}] \in \mathbb{F}^{2 \times 2}$ with $a_{11} \neq 0$. Let $b = a_{21}/a_{11}$. Show that there exists $c \in \mathbb{F}$ such that $A = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ 0 & c \end{bmatrix}$.
2. Suppose $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times k}$. For $1 \leq i \leq m$ and $1 \leq j \leq k$, show that
(a) $(AB)_{i*} = A_{i*}B$ (b) $(AB)_{*j} = AB_{*j}$
Write the results in words.
3. Let $A \in \mathbb{F}^{n \times n}$ be the matrix whose super diagonal entries are all 1, and all other entries are 0. Is it true that $A^n = 0$?
4. Let $A \in \mathbb{C}^{m \times n}$. Is it true that if $A^*A = I$, then $AA^* = I$?
5. Find an elementary matrix E such that $\begin{bmatrix} 2 & 1 & 3 \\ -2 & 4 & 5 \\ 1 & 5 & 9 \end{bmatrix} = E \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 4 \\ -2 & 4 & 5 \end{bmatrix}$.
6. Find an elementary matrix E such that $\begin{bmatrix} 4 & -2 & 1 \\ -2 & 4 & -4 \\ 6 & 1 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -2 & 3 \\ -1 & 4 & 2 \\ 3 & 1 & -2 \end{bmatrix} E$.
7. Compute the RREF of the matrix $\begin{bmatrix} 2 & -1 & -1 & 0 \\ 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & -4 \end{bmatrix}$.
8. Suppose $\text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.
If $A_{*1} = [1 \ 1 \ 3 \ 4]^t$ and $A_{*3} = [2 \ -1 \ 1 \ 3]^t$, then determine A_{*6} .
9. Let $u = [1 \ 2 \ 3 \ 4]$, $v = [2 \ 0 \ 1 \ 1]$, $w = [-3 \ 2 \ 1 \ 2]$ and $x = [1 \ -2 \ -2 \ -3]$.
Construct $y \in \mathbb{R}^{1 \times 4}$ such that $y \neq au + bv + cw + dx$ for any $a, b, c, d \in \mathbb{R}$.
10. Let $A, B \in \mathbb{F}^{2 \times 2}$. Let $E = [e_{ij}]$ with $e_{11} = e_{22} = 0$ and $e_{12} = e_{21} = 1$. Show that if $B = EA$, then $\det(A + B) = \det(A) + \det(B)$.
11. Construct $A, B \in \mathbb{R}^{2 \times 2}$, where $\det(A + B) \neq \det(A) + \det(B)$.
12. For each $n \geq 2$, construct a matrix $A \in \mathbb{F}^{n \times n}$, where no row of A is a scalar multiple of another row but $\det(A) = 0$.

13. Let $A \in \mathbb{F}^{n \times n}$. Is it true that if $\det(A) = 0$, then there exists $i \in \{1, \dots, n\}$ such that $A_{i*} = \sum_{\substack{j=0 \\ j \neq i}}^n a_j A_{j*}$ for some scalars a_j ? What about columns instead of rows?
14. Let $A = \begin{bmatrix} 2 & 1 \\ 6 & 4 \end{bmatrix}$. Express A and A^{-1} as products of elementary matrices.
15. Let $A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$. Use RREF to construct matrices X and Y such that $AX = B$ and $YA = B$.
16. Show that the inverse of an invertible upper triangular matrix is upper triangular.
17. Let $A \in \mathbb{F}^{n \times n}$. Show that if $AB = BA$ for all $B \in \mathbb{F}^{n \times n}$, then there exists $c \in \mathbb{F}$ such that $A = cI$. (Hint: First multiply A with $\text{diag}(1, 2, \dots, n)$. Then, multiply with a matrix whose first row is $[1, 2, \dots, n]$, all diagonal entries are 1, and all other entries are 0.)
18. Let $u, v, w \in \mathbb{C}^5$ be linearly independent vectors. Let α is a complex number. Are the vectors $u + v$, $v + 2w$, $u + \alpha v$, $v - \alpha w$ linearly independent?
19. Prove that nonzero vectors $v_1, \dots, v_n$ are linearly dependent iff one of them is a linear combination of the earlier ones.
20. Using RREF, determine if the following vectors are linearly dependent. If so, then show which vector is a linear combination of others.
 $[1, i, -1, 1 - i]$, $[i, -1, -i, 1 + i]$, $[2, 0, 1, i]$, $[1 + i, 1 - i, -1, -i]$
21. Let $u, v, w \in \mathbb{F}^{4 \times 1}$ and $x = u + 2v$, $y = u - v + 2w$. Determine the RREF of the matrix $\begin{bmatrix} u & v & x & w & y \end{bmatrix} \in \mathbb{F}^{4 \times 5}$.
22. Let $A \in \mathbb{F}^{m \times n}$ and $P \in \mathbb{F}^{m \times m}$. When is $\text{RREF}(PA) = \text{RREF}(A)$?
23. Let $A \in \mathbb{F}^{n \times n}$. Prove that $\text{rank}(A) = 1$ iff there exist vectors $u, v \in \mathbb{F}^{n \times 1}$ such that $A = uv^t$.
24. For each of the following augmented matrices, determine whether the corresponding linear system has (i) no solutions, (ii) a unique solution, and (iii) infinitely many solutions:
 (a) $\left[\begin{array}{ccc|c} 1 & -2 & 4 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$ (b) $\left[\begin{array}{ccc|c} 1 & -2 & 2 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$ (c) $\left[\begin{array}{ccc|c} 1 & -1 & 3 & 8 \\ 0 & 1 & 2 & 7 \\ 0 & 0 & 1 & 2 \end{array} \right]$
25. Show that the linear system $x + y + kz = 1$, $x - y - z = 2$, $2x + y - 2z = 3$ has no solutions for $k = 1$, and has a unique solution for each $k \neq 1$.